Worksheet 5

Time series

Fully worked solutions

Badr ABOUFARASSE, PhD

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1. Core applications

Master the essential definitions and calculations from the course.

Core application 10 min

Exercise 1 – Components and models

1. Match long-run trend, recession, recurring Q4 peak and exceptional shock to T_t, C_t, S_t or R_t.
2. Write the full additive and multiplicative models.
3. Explain when multiplicative seasonality is more plausible than additive seasonality.

Exercise 1 – Solution

$$Y_t = T_t + C_t + S_t + R_t,$$

$$Y_t = T_t \times C_t \times S_t \times R_t.$$

Multiplicative seasonality is more plausible when seasonal amplitude grows with the level of the series.

Distinguishing additive units from multiplicative factors

An additive seasonal component is measured in units of Y and has a constant absolute amplitude. A multiplicative seasonal component is a factor (e.g. 1.20 for +20%), so its absolute effect varies with the trend level. A cycle spans longer fluctuations, whereas an irregular shock is not periodic.

Core application 20 min

Exercise 2 – Moving averages and centring

Quarterly demand over two years is

$$80,\ 100,\ 120,\ 140,\ 88,\ 110,\ 132,\ 154.$$

Describe the main pattern, calculate the five 4-quarter moving averages, centre them for periods 3 to 6, and explain why an even-order moving average must be centred.

Exercise 2 – Detailed solution

$$MA_{2.5} = 110, \quad MA_{3.5} = 112, \quad MA_{4.5} = 114.5, \quad MA_{5.5} = 117.5, \quad MA_{6.5} = 121.$$

$$CMA_3 = 111, \quad CMA_4 = 113.25, \quad CMA_5 = 116, \quad CMA_6 = 119.25.$$

An even-order moving average lies between two periods, so averaging adjacent moving averages centres the estimate on an observed period.

Moving-average alignment table

Window	Sum	MA4	Midpoint
1–4	440	110	2.5
2–5	448	112	3.5
3–6	458	114.5	4.5
4–7	470	117.5	5.5
5–8	484	121	6.5

t	3	4	5	6
CMA4	(110+112)/2=111	(112+114.5)/2=113.25	(114.5+117.5)/2=116	(117.5+121)/2=119.25

Because order 4 is even, the first moving average is centred between t=2 and t=3. Centring gives values at integer dates t=3 through 6.

2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice 25 min

Exercise 3 – Multiplicative decomposition and forecasting

Year	Q1	Q2	Q3	Q4
2005	82	98	120	150
2006	91	111	137	171
2007	104	126	154	194

Given $\widehat{T_t} = 6.343t + 86.939$ and seasonal coefficients $0.789, 0.903, 1.052, 1.255$, interpret the trend slope, check normalization, forecast the four quarters of 2008, and explain multiplicative deseasonalization.

Exercise 3 — Recalculated solution — 2008 forecasts

The trend slope 6.343 implies a 6.343-unit increase per quarter. The supplied seasonal factors sum to 3.999, approximately 4 (rounding).

t	Q	T_t=6.343t+86.939	S'	$T_t \times S'$
13	Q1	169.398	0.789	133.655022
14	Q2	175.741	0.903	158.694123
15	Q3	182.084	1.052	191.552368
16	Q4	188.427	1.255	236.475885

$$\widehat{Y_{13}} = 169.398(0.789) = 133.655022, \quad \widehat{Y_{14}} = 175.741(0.903) = 158.694123.$$

$$\widehat{Y_{15}} = 182.084(1.052) = 191.552368, \quad \widehat{Y_{16}} = 188.427(1.255) = 236.475885.$$

To deseasonalise an observation, divide Y_t by the relevant quarterly factor S'. Retain unrounded figures until the final result.

Assessment / exam practice 65 min

Exercise 5 — Full seasonal decomposition — exam style

A firm observes quarterly sales over three years. Use the multiplicative model $Y_t = T_t \times S_t$, without explicit cycle or irregular component. Do not round intermediate values.

Année / Year	Q1	Q2	Q3	Q4
2019	40	52	48	68
2020	44	58	53	76
2021	50	65	60	84

1. Plot the series and calculate all nine 4-quarter moving averages.
2. Centre the moving averages (t=3 to 10). Build the t, CMA4, t^2, t×CMA4 table.
3. Estimate trend T_t=at+b by OLS on the eight centred moving averages.
4. Calculate Y_t/T_t for all 12 quarters, the four seasonal means and normalised factors summing to 4.
5. Forecast all four quarters of 2022 (t=13 to 16), first deseasonalised, then seasonalised.

Exercise 5 — Full solution — from smoothing to forecasts

1. Four-quarter moving averages:

$$MA_{2.5} = (40 + 52 + 48 + 68)/4 = 52, \quad MA_{3.5} = (52 + 48 + 68 + 44)/4 = 53.$$

Fenêtre / Window	Périodes / Periods	Somme / Sum	MA4	Centre / Midpoint
1	1–4	208	52	2.5
2	2–5	212	53	3.5
3	3–6	218	54.5	4.5
4	4–7	223	55.75	5.5
5	5–8	231	57.75	6.5
6	6–9	237	59.25	7.5
7	7–10	244	61	8.5
8	8–11	251	62.75	9.5
9	9–12	259	64.75	10.5

2. Centring: the nine half-period MA4 values produce eight CMA4 values at integer periods 3 to 10.

$$CMA_3 = (52 + 53)/2 = 52.5, \quad CMA_4 = (53 + 54.5)/2 = 53.75.$$

t	CMA4	t²	t×CMA4
3	52.5	9	157.5
4	53.75	16	215
5	55.125	25	275.625
6	56.75	36	340.5
7	58.5	49	409.5
8	60.125	64	481
9	61.875	81	556.875
10	63.75	100	637.5
Σt=52	462.375	380	3073.5

3. Trend regression using CMA4 values:

$$\bar{t} = 52/8 = 6.5, \quad \overline{CMA} = 462.375/8 = 57.796875.$$

$$V(t) = 380/8 - 6.5^2 = 5.25, \quad \mathrm{Cov}(t, CMA) = 3073.5/8 - 6.5(57.796875) = 8.5078125.$$

$$a = 8.5078125/5.25 = 1.620535714, \quad b = 57.796875 - a(6.5) = 47.263392857.$$

$$\widehat{T_t} = 1.620535714t + 47.263392857.$$

4. Individual seasonal ratios and normalisation:

t	Trimestre / Quarter	Y	T_t	$S_t = Y/T_t$
1	Q1	40	48.8839	0.81826
2	Q2	52	50.5045	1.02961
3	Q3	48	52.125	0.92086
4	Q4	68	53.7455	1.26522
5	Q1	44	55.3661	0.79471

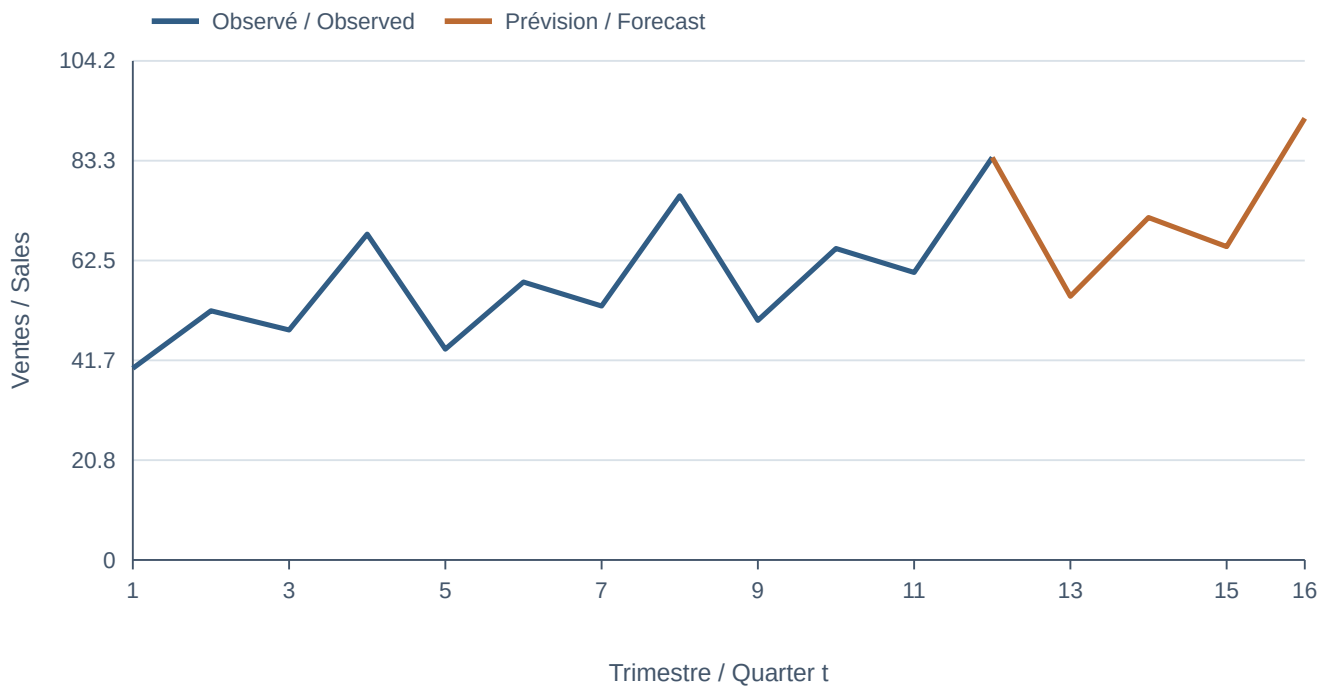
t	Trimestre / Quarter	Y	T_t	$S_t=Y/T_t$
6	Q2	58	56.9866	1.01778
7	Q3	53	58.6071	0.90433
8	Q4	76	60.2277	1.26188
9	Q1	50	61.8482	0.80843
10	Q2	65	63.4688	1.02413
11	Q3	60	65.0893	0.92181
12	Q4	84	66.7098	1.25918

$$\bar{S}_q = \frac{1}{3}\sum_{j=0}^{2}(Y_{4j+q}/\widehat{T}_{4j+q}), \qquad S'_q = \frac{4\bar{S}_q}{\sum_{k=1}^{4}\bar{S}_k}.$$

Trimestre / Quarter	Ratios / Ratios	Moyenne / Mean	S' normalisé / Normalised
Q1	0.81826 ; 0.79471 ; 0.80843	0.807135	0.805376
Q2	1.02961 ; 1.01778 ; 1.02413	1.02384	1.021609
Q3	0.92086 ; 0.90433 ; 0.92181	0.915667	0.913671
Q4	1.26522 ; 1.26188 ; 1.25918	1.262095	1.259344
TOTAL	—	4.008737	4

5. Forecast: T_t is the deseasonalised forecast; multiply by the relevant quarterly factor S' for the final forecast.

t	Trimestre / Quarter	T_t	S'	Prévision / Forecast
13	Q1	68.3304	0.805376	55.0316
14	Q2	69.9509	1.021609	71.4624
15	Q3	71.5714	0.913671	65.3927
16	Q4	73.192	1.259344	92.1739



Ventes observées et prévisions 2022 / Observed and 2022 forecasts

Check: the four normalised seasonal factors sum to 4. The 2022 forecasts are approximately 55.03, 71.46, 65.39 and 92.17.

3. Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice 15 min

Exercise 4 – Forecast evaluation

For this exercise, suppose actual 2008 values were $\mathbf{136, 162, 188, 245}$. Using the Exercise 3 forecasts, calculate forecast errors, MAE and RMSE, and explain the value of a separate validation period and a naive benchmark.

Exercise 4 — Corrected solution — 2008 validation

Forecasts are exactly those computed in exercise 3, without substituting rounded figures from another source.

Q	Y	Prévision / Forecast	e=Y−Ŷ	|e|	e²
1	136	133.655022	2.344978	2.344978	5.49892
2	162	158.694123	3.305877	3.305877	10.92882
3	188	191.552368	-3.552368	3.552368	12.61932
4	245	236.475885	8.524115	8.524115	72.66054

$$MAE = (2.344978 + 3.305877 + 3.552368 + 8.524115)/4 = 4.431835.$$

$$RMSE = \sqrt{(2.344978^2 + 3.305877^2 + (-3.552368)^2 + 8.524115^2)/4} = 5.042509.$$

$$ME = (2.344978 + 3.305877 - 3.552368 + 8.524115)/4 = 2.655651.$$

The model underpredicts on average. Validation should be out of sample; a naive forecast is a useful benchmark.

Advanced practice 20 min

Extra 1 — Forecast validation and structural break

Actual monthly demand is 100, 104, 108, 150, 154 and 158. A trend model fitted on the first three months predicts 100, 104, 108, 112, 116 and 120.

1. Calculate forecast errors for months 4–6.
2. Calculate MAE and mean error over the validation period.
3. Diagnose the model.
4. Propose a responsible forecasting response.

Use error = actual − forecast. A persistently positive error signals systematic underforecasting.

Extra 1 — Forecast validation and structural break

Detailed solution

Errors. Month 4: 150−112 = 38; month 5: 154−116 = 38; month 6: 158−120 = 38.

Validation metrics. MAE = (38+38+38)/3 = **38**. Mean error is also **+38**, showing systematic underforecasting rather than random noise.

Diagnosis. The level jumps between months 3 and 4 while the subsequent slope remains close to four units per month. This is consistent with a level structural break. A model trained only before the break is obsolete.

Response. Investigate the business event causing the jump, re-estimate the level using post-break data, compare with a naive benchmark, retain a validation window, and issue prediction intervals. Do not simply average pre- and post-break observations without explaining the regime change.